

XAI-Driven Modality Bias Detection and Mitigation in Multimodal Deep Learning Models

Deep Learning — Semester Project Proposal
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1. Problem Description

Multimodal deep learning models that jointly process vision and language inputs have demonstrated remarkable capabilities, yet they suffer from a critical and often invisible flaw: **modality bias**. These models may disproportionately rely on one modality—typically language—while underutilizing the other, leading to unreliable predictions, poor generalization, and reduced trustworthiness. Recent studies report strong language dominance in large Vision-Language Models (VLMs), with more than 56% of predictions matching text-only answers compared with only 27% matching image-only inputs. This project investigates whether XAI methods can expose this imbalance and whether the resulting attribution signals can be used to mitigate it during fine-tuning.

2. Objectives

1. Detect and quantify modality bias in a pretrained VLM (PaliGemma-3B) using XAI attribution methods, specifically attention-weight analysis and Integrated Gradients, to produce a measurable *Modality Bias Score* per sample and across datasets.
2. Evaluate whether XAI attribution scores can mitigate modality bias by using them as a feedback signal in a bias-regularization loss during fine-tuning, inspired by the Reveal-to-Revise framework.
3. Develop a novel **Balanced Modal Attention (BMA)** mechanism—a lightweight, learnable reweighting layer that dynamically equalizes the contribution of visual and textual tokens within attention layers without degrading task performance.
4. Benchmark and validate the proposed approach on standard multimodal datasets such as VQAv2 and COCO Captions, comparing bias scores and task performance before and after intervention.

3. Proposed Methodology

Step 1 — Baseline Bias Measurement: Using PaliGemma-3B attention outputs, define the Modality Bias Score (MBS) as the proportion of attention mass assigned to image tokens relative to total attention mass:

$$\text{MBS} = \Sigma(\text{attention on image tokens}) / \Sigma(\text{all attention weights})$$

An MBS near 0 indicates language dominance, while an MBS near 1 indicates vision dominance. A balanced model should trend toward 0.5.

Step 2 — XAI Attribution using Integrated Gradients: Apply Integrated Gradients to the fused token embeddings to estimate input-level attribution for each modality. These attribution scores will be compared with attention-based measurements.

Step 3 — Balanced Modal Attention (BMA): Introduce a lightweight learnable gating mechanism into the attention computation:

$$\text{Attn}_{\text{balanced}} = \alpha \cdot \text{Attn}_{\text{image}} + (1 - \alpha) \cdot \text{Attn}_{\text{text}}$$

The gate α is trained together with a bias-regularization term $\mathbf{R}_{\text{bias}} = (\text{MBS} - 0.5)^2$, encouraging balanced contribution from the two modalities while preserving the original task objective.

Step 4 — Evaluation: Evaluate VQA accuracy, captioning quality (including BLEU), MBS distributions, and per-group performance on COCO/VQAv2.

4. Plan of Action

Phase	Task
1	Literature review; baseline setup; load PaliGemma-3B on COCO/VQAv2
2	Implement bias detection: extract attention weights, compute image/text attention ratios, and generate heatmaps
3	Implement Integrated Gradients via Captum; compare attribution scores; quantify bias on 500+ samples
4	Design and integrate BMA; implement bias-regularization loss R_bias
5	Fine-tune with BMA + regularization; evaluate accuracy versus bias score
6	Analysis, ablations, visualizations, and report writing

5. Expected Outcomes

- A working modality-bias detection pipeline with quantitative metrics and XAI visualizations.
- Empirical evidence on whether XAI attribution scores can serve as an effective bias-mitigation signal.
- A novel, plug-and-play Balanced Modal Attention layer that reduces modality dominance.
- A comparative analysis demonstrating the accuracy–fairness trade-off with and without bias correction.

6. Tools & Technology Stack

Python · PyTorch · HuggingFace Transformers · Captum (XAI) · COCO / VQAv2 · Google Colab (T4/A100 GPU) · PaliGemma-3B-pt-224

7. References

1. A Multimodal XAI Framework for Trustworthy CNNs — arXiv:2510.12957.
2. MLLMs are Deeply Affected by Modality Bias — arXiv:2505.18657.
3. The Multimodal Paradox — arXiv:2505.03020.